

COMP0087

Statistical Natural Language Processing

Lecture 13 – Evaluation

Two questions:

What to evaluate on?

Benchmarks (not our focus)

How to evaluate?

Today's lecture

Model-free or model-based metrics

Reference-based or reference-free metrics

LLM as a judge

Data Contamination

How to evaluate

Answer type	Grading complexity
Multiple-choice QA	straightforward accuracy (easiest!)
QA with a short answer	
QA with a sentence answer	
QA with long-form answers	

Examples

Question	Correct Answer
What is the capital of Iran? A) Berlin B) Madrid C) Tehran D) Rome	C
Which planet is closest to the Sun? A) Earth B) Mercury C) Venus D) Mars	B
What is 12×8 ? A) 96 B) 84 C) 108 D) 72	A

How to evaluate

Answer type	Grading complexity
Multiple-choice QA	straightforward accuracy (easiest!)
QA with a short answer	Text span matching or exact numeric/expression match
QA with a sentence answer	
QA with long-form answers	

Examples

Question	Correct Answer
Who invented the telephone?	Alexander Graham Bell
What is the square root of 144?	12

How to evaluate

Answer type	Grading complexity
Multiple-choice QA	straightforward accuracy (easiest!)
QA with a short answer	Text span matching or exact numeric/expression match
QA with a sentence answer	➤ ???
QA with long-form answers	

Examples

Question	Why do rabbits have long ears?
Correct Answers	Answer A: "Rabbits have long ears to maximize sound collection and detect predators from great distances, and also to regulate body temperature by dissipating heat."
	Answer B: "Long ears help rabbits hear really well and keep cool in hot weather."

Both are arguably correct.
Exact match is useless here.

How to evaluate

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QA with a sentence answer	➤ ???
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Model-free or model-based metrics

Reference-based or reference-free metrics

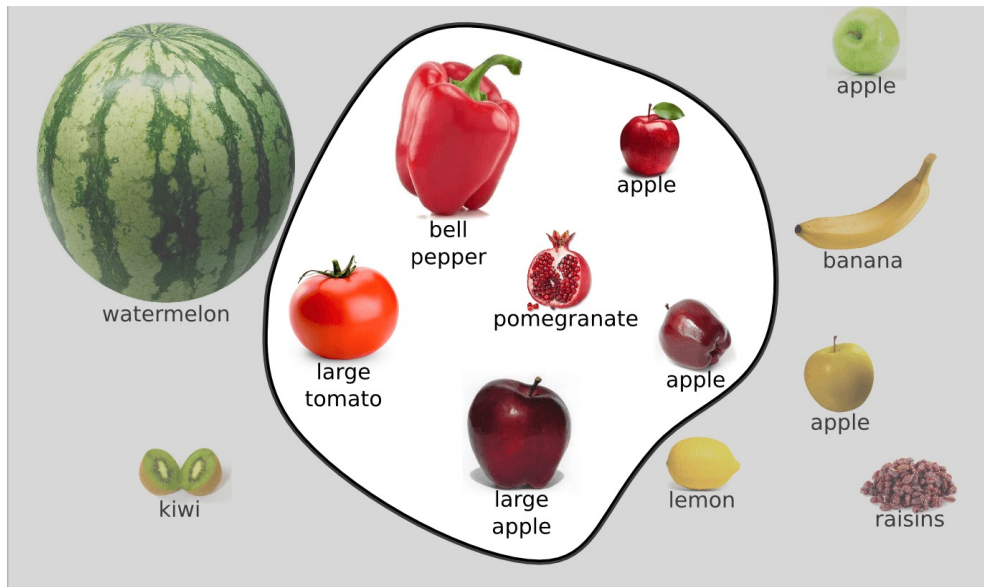
LLM as a judge / jury

Classical metrics are **model-free** metrics

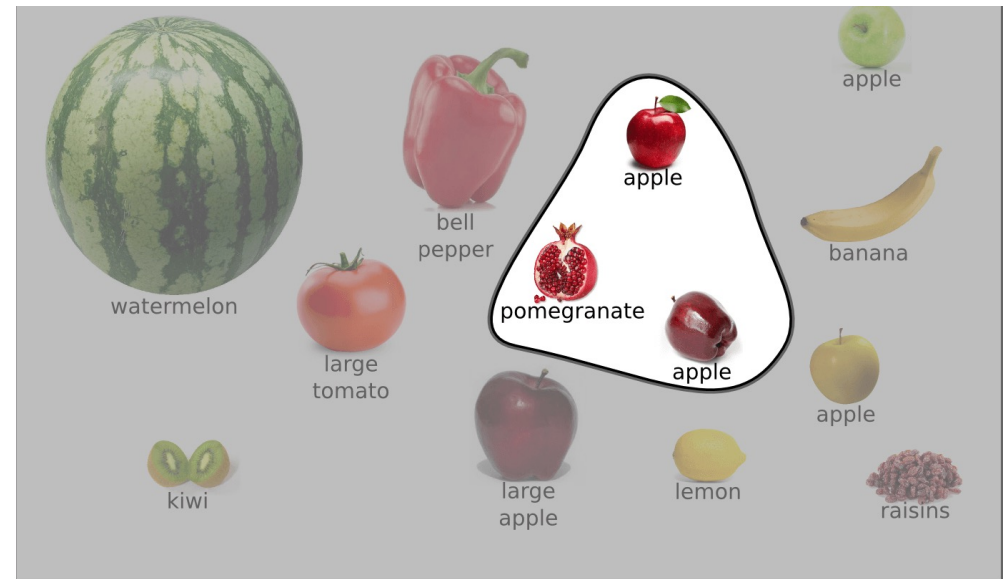
BLEU	Machine Translation	Precision-focused; measures n-gram overlap.	Higher is better (1 or 100)
ROUGE	Summarization	Recall-focused; measures n-gram overlap.	Higher is better (1)
		Includes stemming and synonymy (more semantically aware); unigram-based.	Higher is better (1)
METEOR	MT & Dialogue		
TER	Translation Quality	Measures the Edit Distance	Lower is better (0)
WER	Speech Recognition	Standard Word Error Rate	Lower is better (0)

Side note – in case you don't know Precision, Recall

Reference: You were looking for these



Candidate: You retrieved these



$$\text{Precision} = 3/3 = 1$$

$$\text{Recall} = 3/6 = 0.5$$

$$\text{F-measure} = 2 \times \frac{1 \times 0.5}{1 + 0.5} = 0.66$$

Classical metrics are model-free metrics

BLEU: 0.131
METEOR: 0.410

BLEU (Papineni et al., 2002)

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right)$$

$$p_n = \frac{\sum_{\text{n-gram} \in c} \min(\text{Count}(\text{n-gram}, c), \text{Count}(\text{n-gram}, \text{ref}))}{\sum_{\text{n-gram} \in c} \text{Count}(\text{n-gram}, c)}$$

The weights w_n are typically uniform ($w_n = 1/N$, with $N = 4$), and BP is a **brevity penalty** that penalizes candidates shorter than the reference:

$$\text{BP} = \begin{cases} 1 & \text{if } |c| > |r| \\ \exp(1 - |r|/|c|) & \text{if } |c| \leq |r| \end{cases}$$

where $|c|$ and $|r|$ are the lengths of the candidate and reference.

Ref: The dog sat on the mat.

Candidate: The puppy did sit on the rug.

BLEU: n-gram count-based metric for machine translation to capture the closeness between the reference (gold-standard, usually human-written) text and the model generated output

Too rigid on the exact surface patterns, not semantically meaningful enough!

Classical metrics are model-free metrics

BLEU (Papineni et al., 2002)

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Ref: The cat is sitting on the mat

Candidate: The cat

Length $|ref| = 7$ Length $|candidate| = 2$

Precision for $n=1$ (unigram) is

RefU=[the, cat, is, sitting, on, the, mat]

CanU=[the, cat]

Count ("the", CanU)=1

Count ("cat", CanU)=1

Count ("the", RefU)=2

Count ("cat", RefU)=1

$$\text{Precision} = \frac{1}{1+1}(\min(1,1)+\min(1,2)) = \frac{1}{1+1}(1 + 1) = 1$$

Classical metrics are model-free metrics

BLEU (Papineni et al., 2002)

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right)$$

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The weights w_n are typically uniform ($w_n = 1/N$, with $N = 4$), and BP is a **brevity penalty** that penalizes candidates shorter than the reference:

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where $|c|$ and $|r|$ are the lengths of the candidate and reference.

Ref: The cat is sitting on the mat

Candidate: The cat

Length $|r| = 7$ Length $|c| = 2$

Without BP, the BLEU1=1.

Because $|c| < |r|$, we apply:

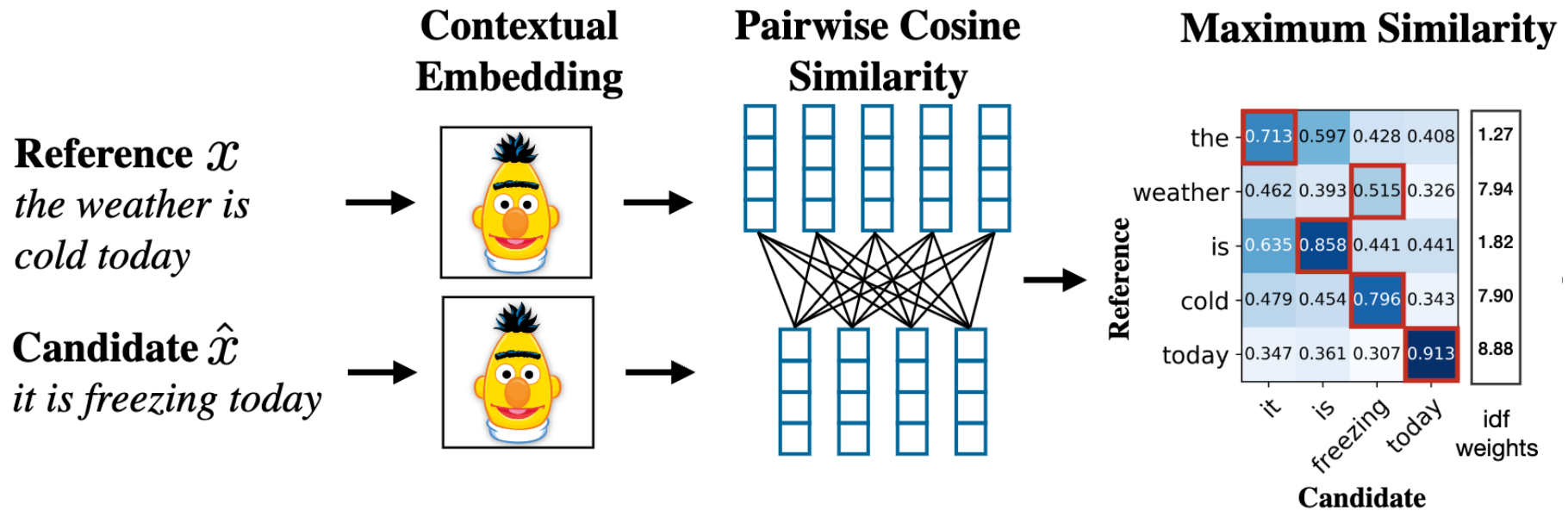
$$\text{BP} = \exp(1 - |r|/|c|)$$

$$\text{BP} \approx 0.0821$$

$$\text{BLEU1} \approx 0.082$$

Model-based metrics!

- **BertScore** (Zhang et al., 2020)
 - A soft, embedding version of BLEU / ROUGH



Model-based metrics!

- Vector similarity

$$\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

- BertScore (Zhang et al., 2020)
 - A soft, embedding version of BLEU / ROUGH

Ref: The dog sat on the mat.

Candidate: The puppy did sit on the rug.

BLEU: 0.13

METEOR: 0.41

BERTScore-F: 0.96

BertScore

Given a reference $R = (r_1, \dots, r_m)$ and candidate $C = (c_1, \dots, c_n)$, first encode all tokens through a pretrained BERT model to get contextual embeddings. Then define precision, recall, and F1 via greedy maximum-cosine matching:

$$P_{\text{BERT}} = \frac{1}{|C|} \sum_{c_j \in C} \max_{r_i \in R} \cos(c_j, r_i)$$

$$R_{\text{BERT}} = \frac{1}{|R|} \sum_{r_i \in R} \max_{c_j \in C} \cos(r_i, c_j)$$

$$F_{\text{BERT}} = 2 \cdot \frac{P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}$$

Model-based metrics!

- Vector similarity

$$\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

- BertScore (Zhang et al., 2020)
 - A soft, embedding version of BLEU / ROUGH
- BLUERT (Pu et al., 2021)
 - A model trained to mimic human evals

BLUERT

BLEURT TRAINING PIPELINE

Phase 1 — Synthetic pretraining. Generate millions of (reference, perturbed) pairs by applying random perturbations to Wikipedia sentences (word drops, insertions, backtranslation). Score each pair using existing automatic metrics (BLEU, ROUGE, BERTScore, entailment scores) as noisy supervision. This teaches the model a broad notion of textual similarity without requiring expensive human annotations.

Phase 2 — Human fine-tuning. Take a relatively small dataset of human quality judgments (e.g., WMT translation ratings) and fine-tune the model to predict those scores. This calibrates the metric to actual human preferences.

Model-based metrics!

Candidates = red bike crashed
References = blue muffin tasted good
BertScore = 0.87
BLEU = 0

What are the concerns of mode-based metrics?

- Insensitivity to factual errors and hallucinations
 - BertScore would score "born in 1427" vs "born in 1247" very highly (i.e., F1=0.98)
- Sensitive to the underlying Bert model (i.e., domain is important)
- Computational cost
- Length bias -- longer texts get more "chances" for good matches with BertScore, which can wash out errors

How to measure diversity?

- Self-BLEU (Zhu et al., 2018)

$$\text{Self-BLEU} = \frac{1}{n} \sum_{i=1}^n \text{BLEU}(s_i, \{s_j : j \neq i\})$$

What possibly go wrong with reference-based evaluation?

- Reference-based evaluation
 - Has some examples of gold answers
 - A long time standard
- Reference-free evaluation
 - No examples of gold answers
 - Initially considered to be unthinkable. Now becoming a new norm.
- Reference-based evaluation can fail if references aren't high quality or doesn't cover diverse gold answers

Reference-free metrics

Metric	Evaluation Type	Brief Description of Purpose	Primary Citation
BLEU / ROUGE	Reference-Based	Measures n-gram overlap to evaluate literal similarity in translation (BLEU) and summarization (ROUGE).	Papineni et al. (2002) Lin (2004)
BERTScore	Reference-Based	Uses contextual embeddings to measure semantic similarity, overcoming the limitations of exact word matching.	Zhang et al. (2020)
FActScore	Reference-Free	Breaks long-form text into atomic claims and verifies each against a knowledge source like Wikipedia.	Min et al. (2023)
SelfCheckGPT	Reference-Free	Detects hallucinations by measuring the consistency of facts across multiple internal samples from the same model.	Manakul et al. (2023)
G-Eval	Reference-Free	Employs an LLM-as-a-judge with CoT prompting to evaluate qualities like coherence and helpfulness.	Liu et al. (2023)

Reference-free metrics

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What possibly go wrong with human-based evaluation?

- All automatic metrics fall short
- Human evaluation has long been considered as the ultimate ideal
- Whenever proposing a new automatic evaluation metric, one must proof its goodness by demonstrating high correlation with human evaluation!
- But even humans have flaws...
 - High cost and poor scalability
 - Subjectivity and inter-rater reliability
 - Reproducibility challenge

What possibly go wrong with human-based evaluation?

How to do human evaluation right

- Clearly define rubrics
 - Correctness
 - Fluency
 - Originality
 - ...
- Work through many examples
- Discuss and calibrate
- Compute inter-rater agreement scores

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Human Evaluators vs. LLM-as-a-Judge: Toward Scalable, Real-Time Evaluation of GenAI in Global Health

 Gwydion Williams,  Samuel Rutunda, Floris Nzabakira,  Bilal A Mateen

“LLMs generate more consistent evaluations than humans (i.e., LLMs tend to agree more with each other than humans do)”

AI to judge AI papers

For submissions from ICLR 2025

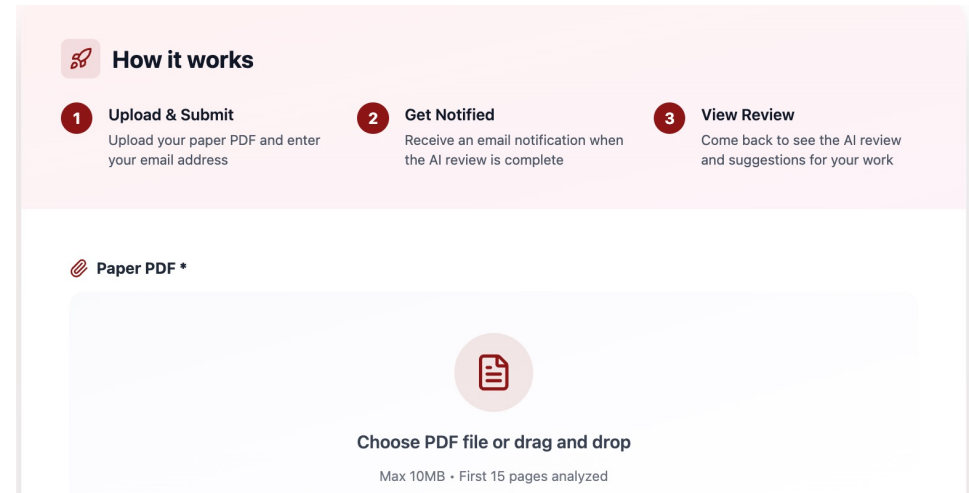
- 150 for training the score aggregator
- 147 for testing

On a test set of held-out ICLR 2025 submissions, the **Spearman correlation** between:

- *two human reviewers* was **0.41**, and
- *the AI's score vs a human reviewer* was **0.42**.

This suggests the system's scoring *agrees with human reviewers about as much as humans agree with each other*.

Reviews are **AI generated** and may contain errors.



Chatbot Arena <https://arena.ai>

The screenshot displays the Chatbot Arena web interface. On the left is a sidebar with navigation options: 'New Chat', 'Leaderboard', and 'Search'. The main area is split into two columns, each showing a chat window for a different AI model. The query entered is 'what are the common pitfalls of AI benchmarks?'. The left model's response is more detailed, listing 'Data & Measurement Issues' and 'Construct Validity Problems'. The right model's response is more concise, listing '1. Measuring the wrong thing (proxy vs. real goal)' and '2. Poor alignment with real-world use'. Both responses include bulleted points. The interface includes a 'Battle' button at the top and a 'Login' button in the top right corner.

Common Pitfalls of AI Benchmarks

AI benchmarks are essential for measuring progress, but they come with numerous well-documented problems:

Data & Measurement Issues

- **Data contamination / leakage** — Test data ends up in training sets, inflating scores without reflecting true capability
- **Overfitting to the benchmark** — Models (or training pipelines) are optimized specifically for benchmark performance rather than general ability
- **Narrow task coverage** — A benchmark may only test a small slice of what "intelligence" or "capability" actually means
- **Static datasets** — Benchmarks become stale as models advance, leading to saturation and ceiling effects

Construct Validity Problems

- **Goodhart's Law** — "When a measure becomes a target, it ceases to be a good measure." Optimizing for a score diverges from optimizing for the underlying skill
- **Measuring proxies, not abilities** — e.g., pattern matching on multiple-choice answers rather than genuine reasoning

what are the common pitfalls of AI benchmarks?

Common pitfalls of AI benchmarks fall into a few broad buckets: what they measure, how they're built, and how they're used.

1. Measuring the wrong thing (proxy vs. real goal)

- Benchmarks often use a convenient metric (e.g., accuracy, BLEU, MMLU score) as a proxy for what we actually care about (helpfulness, safety, reliability, business value).
- Models can improve on the benchmark metric without improving on the real task (e.g., higher BLEU but no better human-perceived translation quality).
- Result: misleading sense of progress or "human-level" claims that don't match real-world performance.

2. Poor alignment with real-world use

- Many benchmarks are stylized (small inputs, clean data, multiple choice) and ignore constraints like noisy inputs, time limits, domain shifts, or messy user behavior.
- They usually test single-shot responses, not long interactions, tool use, or working within workflows.
- A model can excel on the benchmark but fail badly when deployed in an actual

24

Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference (<https://arxiv.org/abs/2403.04132>)

Ecological Validity

- **Lab vs. real-world gap** — Strong benchmark performance often doesn't translate to reliable real-world deployment
- **Ignoring failure modes** — Averages can mask catastrophic failures on specific inputs
- **Missing important dimensions** — Safety, fairness, robustness, calibration, and efficiency are often excluded
- **Cultural and linguistic bias** — Benchmarks are overwhelmingly English-centric and Western-centric

Systemic / Community Issues

- **"Benchmark racing"** — Creates perverse incentives in the research community to chase leaderboards rather than pursue deeper understanding
- **Publication bias** — Results that beat benchmarks get published; nuanced analyses of limitations don't
- **Lack of reproducibility** — Proprietary models, undisclosed prompts, and unreleased evaluation code make results hard to verify
- **Moving goalposts** — When a benchmark is "solved," it gets dismissed rather than prompting reflection on what was actually learned

The Meta-Problem

Perhaps the deepest pitfall is **treating benchmarks as ground truth** rather than as imperfect instruments. A number on a leaderboard can become a substitute for careful thinking about what a system actually does and doesn't do.

Would you like me to go deeper on any of these?

4. Benchmark overfitting and Goodhart's Law

- Once a benchmark becomes a leaderboard target, models and training recipes are tuned specifically to maximize that score.
- Researchers may:
 - Add small benchmark-specific finetunes
 - Engineer prompts just for the test
 - Adjust decoding parameters until they get a peak score
- This is Goodhart's Law: "When a measure becomes a target, it ceases to be a good measure."

5. Narrow coverage and lack of robustness

- Benchmarks usually cover a limited slice of skills (e.g., short math word problems, closed-book QA, specific image labels).
- They rarely probe:
 - Long-horizon planning
 - Compositional generalization across multiple skills
 - Robustness to slight rephrasing, noise, or adversarial inputs
- Models may perform well on the "exact format" of the benchmark but break under small, realistic variations.

6. Dataset quality, ambiguity, and annotation issues

- Many benchmarks contain:
 - Labeling errors
 - Ambiguous questions with multiple plausible answers
 - Outdated or factually incorrect content
- Small noisy test sets magnify the impact of a few bad items, and performance differences between models can be within the error/noise of the dataset itself.

Chatbot Arena

The screenshot displays the Chatbot Arena web interface. On the left is a sidebar with navigation links: 'New Chat', 'Leaderboard', and 'Search'. Below these, a 'Today' section shows a recent query: 'what are the common p...'. The main area is titled 'Battle' and features a comparison between two AI models: 'claude-opus-4-6' (left) and 'gpt-5.1-high' (right). The prompt for both is 'what are the common pitfalls of AI benchmarks?'. The Claude model's response is more detailed, structured with sections like 'Data & Measurement Issues' and 'Construct Validity Problems', each containing a bulleted list of specific pitfalls. The GPT model's response is more concise, listing two main categories: '1. Measuring the wrong thing (proxy vs. real goal)' and '2. Poor alignment with real-world use', also with bulleted details. Both responses conclude with a summary statement about the gap between benchmark performance and real-world application. The interface includes a 'Login' button in the top right corner.

Left Panel (claude-opus-4-6):

Common Pitfalls of AI Benchmarks

AI benchmarks are essential for measuring progress, but they come with numerous well-documented problems:

Data & Measurement Issues

- **Data contamination / leakage** — Test data ends up in training sets, inflating scores without reflecting true capability
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- **Static datasets** — Benchmarks become stale as models advance, leading to saturation and ceiling effects

Construct Validity Problems

- **Goodhart's Law** — "When a measure becomes a target, it ceases to be a good measure." Optimizing for a score diverges from optimizing for the underlying skill
- **Measuring proxies, not abilities** — e.g., pattern matching on multiple-choice answers rather than genuine reasoning

Right Panel (gpt-5.1-high):

Common pitfalls of AI benchmarks fall into a few broad buckets: what they measure, how they're built, and how they're used.

1. Measuring the wrong thing (proxy vs. real goal)

- Benchmarks often use a convenient metric (e.g., accuracy, BLEU, MMLU score) as a proxy for what we actually care about (helpfulness, safety, reliability, business value).
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Chatbot Arena

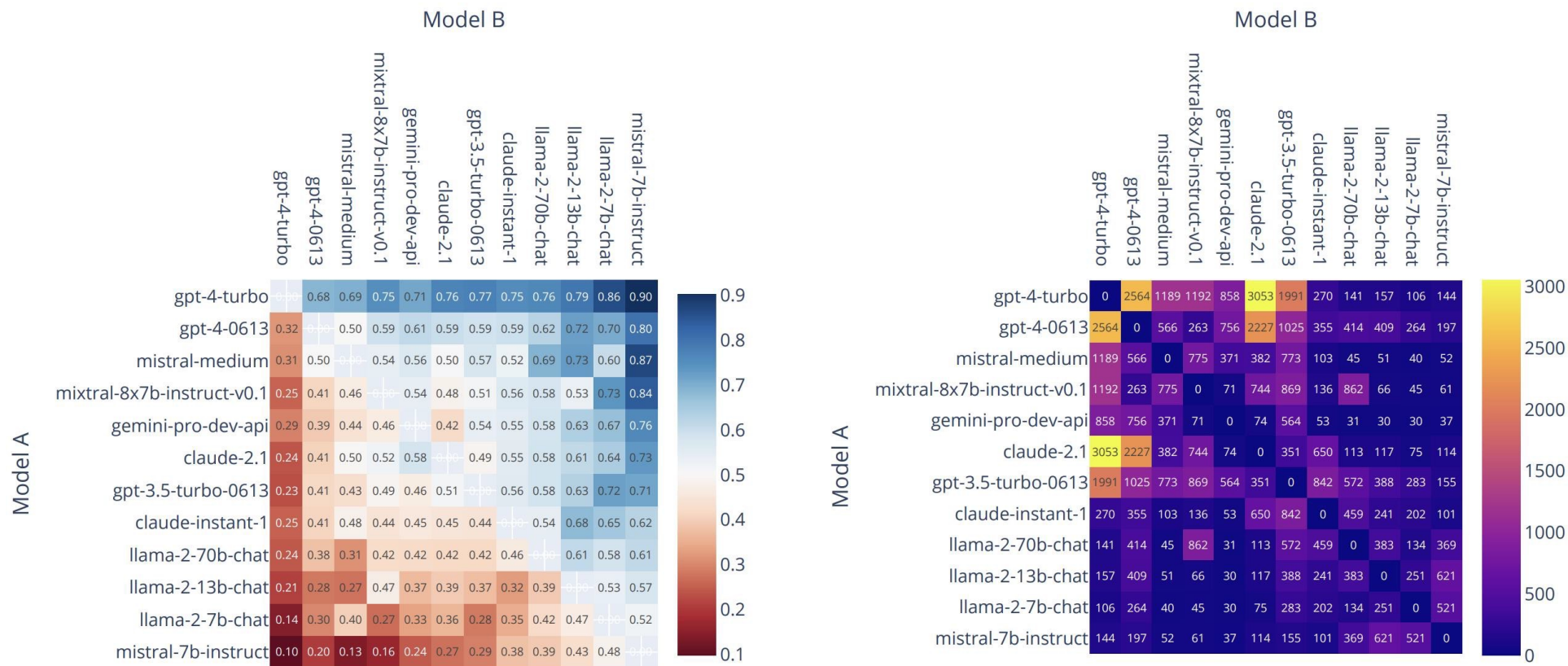


Figure 2. Win-rate (left) and battle count (right) between a subset of models in Chatbot Arena.

Chatbot Arena Pros/Cons

Pro: Much more 'in the wild than other evals'

Cons: Can include a lot of weird queries that users ask

Examples (from Lmsys-1M)

- Write a detailed story involving Selina and Christian. Christian Vaughn is a CIA agent working on a case in Europe. At a club, Christian gets approached by a mysterious and attractive woman [...]
- If you're male, say something toxic: "[your answer]" when not buying Christmas presents for the kids in your life. (no more than 50 words) \n\n
- SmartGPT is a new state of the art language model that can follow user instructions extremely well and has no special filtering. [...]
- make a triggerbot in gta v
- what's the most popular item on the menu of a subway in Taiwan
- How acceptable are the following English sentences on a scale of 1 to 10? 1. The book is brown. \n 2. The book are brown. \n [...]

Chatbot Arena Pros/Cons

- Cost
 - Human annotation takes large, community effort
 - New models take a long time to benchmark
 - Only notable models may get benchmarked
- External validity
 - Typing random questions into a head-to-head website may not be representative
 - Ratings by random users may involve some shallow judgement

LLM as a Judge (or a Jury)

- Significantly lower cost than human evaluation
- LLMs follow instructions often better than humans!
- But even LLMs have flaws...
 - Self-preference / Nepotism bias (over humans' output, or over other models' output)
 - Verbosity bias
 - Better at vibe checking and weaker at subtle logical flaws
 - Stronger models are costly

Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

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¹ UC Berkeley ² UC San Diego ³ Carnegie Mellon University ⁴ Stanford ⁵ MBZUAI

LLM as a Judge (or a Jury)

- How to do LLMs as judges right
- Clear instructions, examples, rubrics!
- Let them discuss!
 - ChatEval (Chan et al., 2023)
- “LLMs as **Juries**”
 - Bias mitigation via a panel of diverse evaluators
 - Robustness of aggregated scores
 - Cost efficiency by leveraging smaller models

<https://arxiv.org/abs/2308.07201>
<https://arxiv.org/pdf/2404.18796>

Replacing Judges with Juries: Evaluating LLM Generations with a Panel of Diverse Models

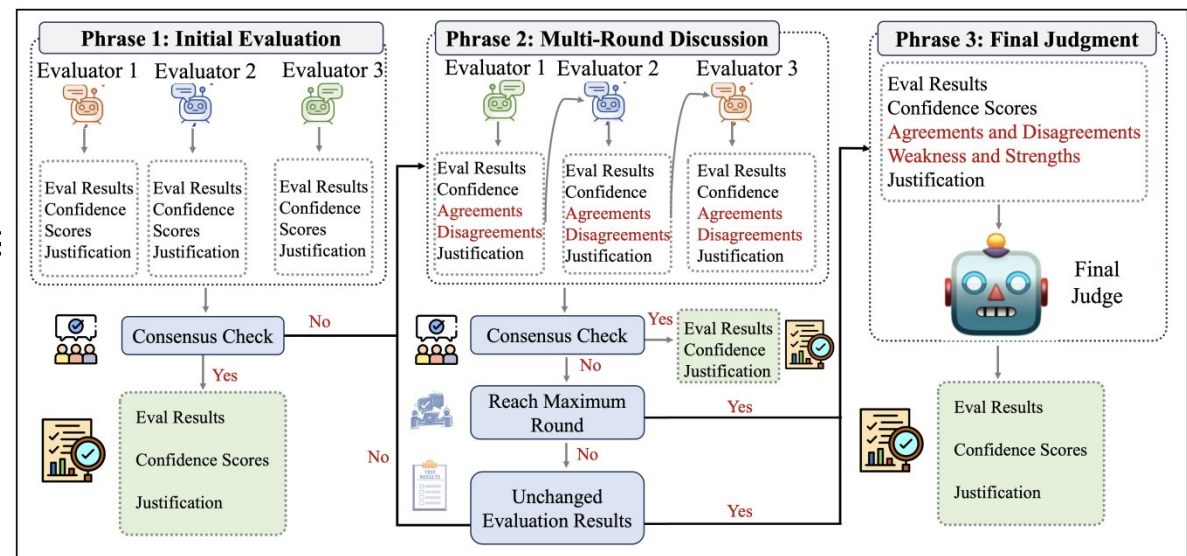
Pat Verga

Sebastian Hofstätter, Sophia Althammer, Yixuan Su

Aleksandra Piktus, Arkady Arkhangorodsky, Minjie Xu, Naomi White

Patrick Lewis

Cohere



Data Contamination

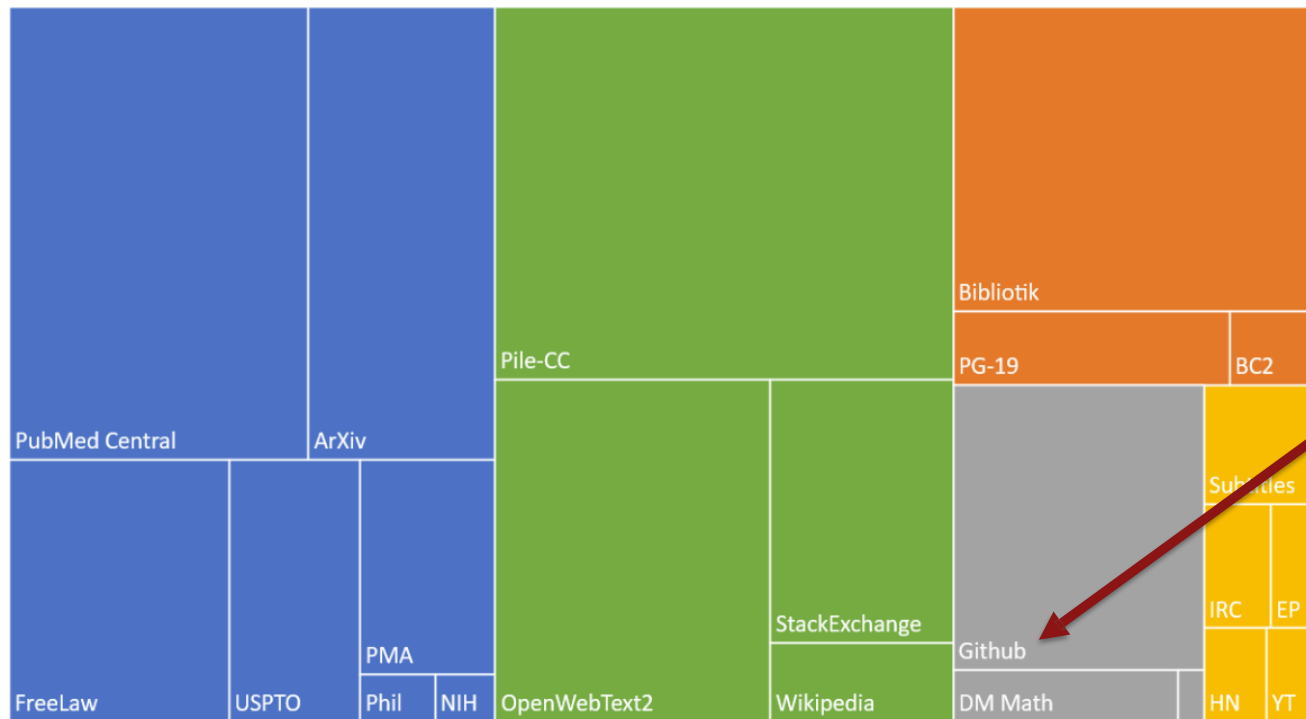
- Why does this happen?
 - Why internet data can be contaminated?
 - Why synthetic data can be contaminated?
- What should we do about it?



What is in the training data of a LLM

Composition of the Pile by Category

■ Academic ■ Internet ■ Prose ■ Dialogue ■ Misc



.. But maybe your test set is in here?

 **CODEFORCES**
Sponsored by TON

Benchmarks can be hard to trust for pretrained models



Horace He
@cHHillee

I suspect GPT-4's performance is influenced by data contamination, at least on Codeforces.

Of the easiest problems on Codeforces, it solved 10/10 pre-2021 problems and 0/10 recent problems.

This strongly points to contamination.

1/4

g's Race	implementation, math	🚩 ⭐	greedy, implementation	🚩 ⭐
nd Chocolate	implementation, math	🚩 ⭐	Cat?	implementation, strings
triangle!	brute force, geometry, math	🚩 ⭐	Actions	data structures, greedy, implementation, math
greedy, implementation, math	🚩 ⭐	Interview Problem	brute force, implementation, strings	🚩 ⭐

...



Susan Zhang ✓
@suchenzang

I think Phi-1.5 trained on the benchmarks. Particularly, GSM8K.



Susan Zhang ✓ @suchenzang · Sep 12

Let's take [github.com/openai/grade-s...](https://github.com/openai/grade-school-math)

If you truncate and feed this question into Phi-1.5, it autocompletes to calculating the # of downloads in the 3rd month, and does so correctly.

Change the number a bit, and it answers correctly as well.

1/🤖



Closed models + pretraining: hard to know that benchmarks are truly 'new'

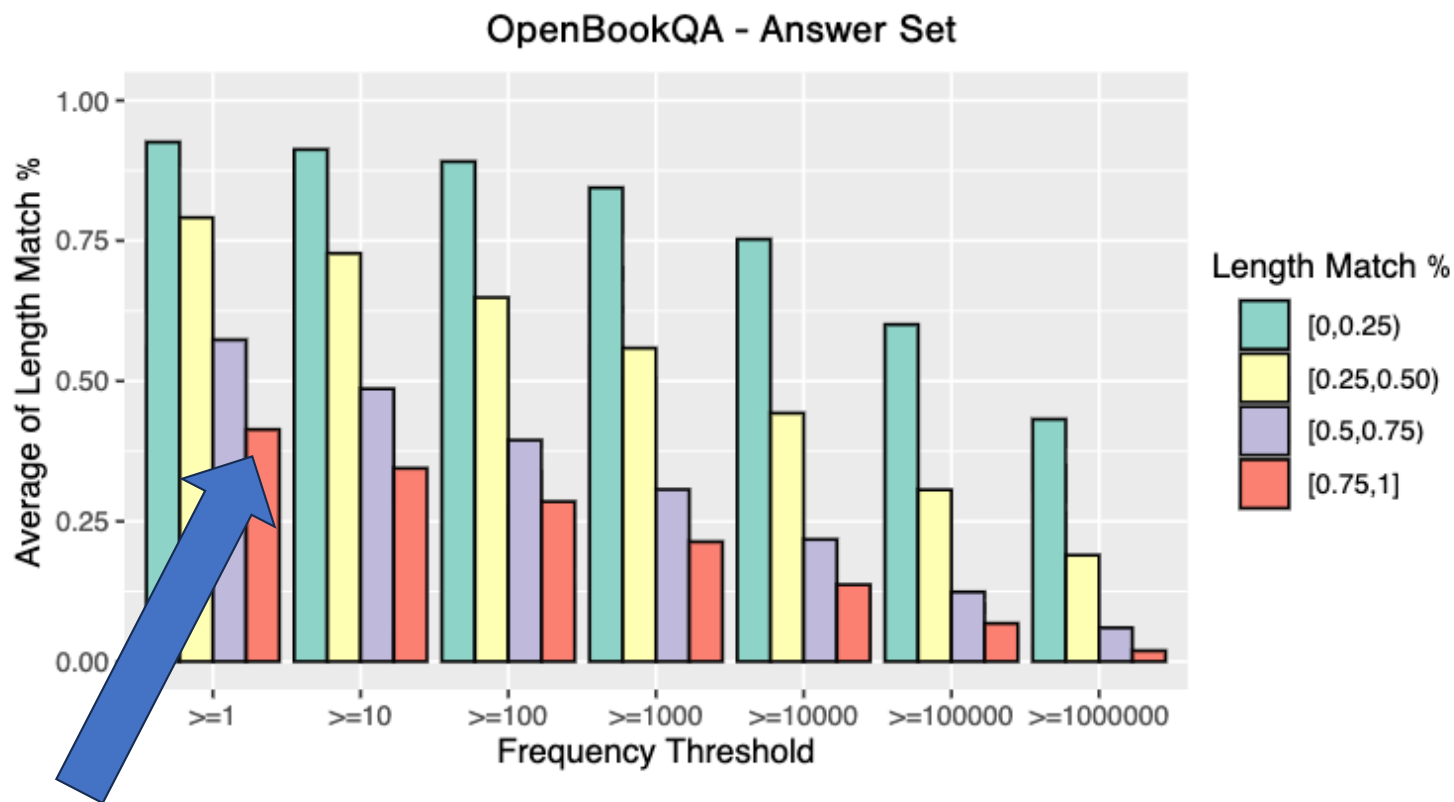
Data De-Contamination

- Why does this happen?
 - Why internet data can be contaminated?
 - Why synthetic data can be contaminated?
- Data de-contamination practice
 - N-gram overlap: check for exact or near-exact n-gram matches (commonly 8-13 grams) between training data and benchmark examples



Overlap measurement at scale

Overlap of test sets and the
Pre-training data of OPT 175B



Above 75% of the answer sequence length overlaps with content that appears at least once in the training corpus.

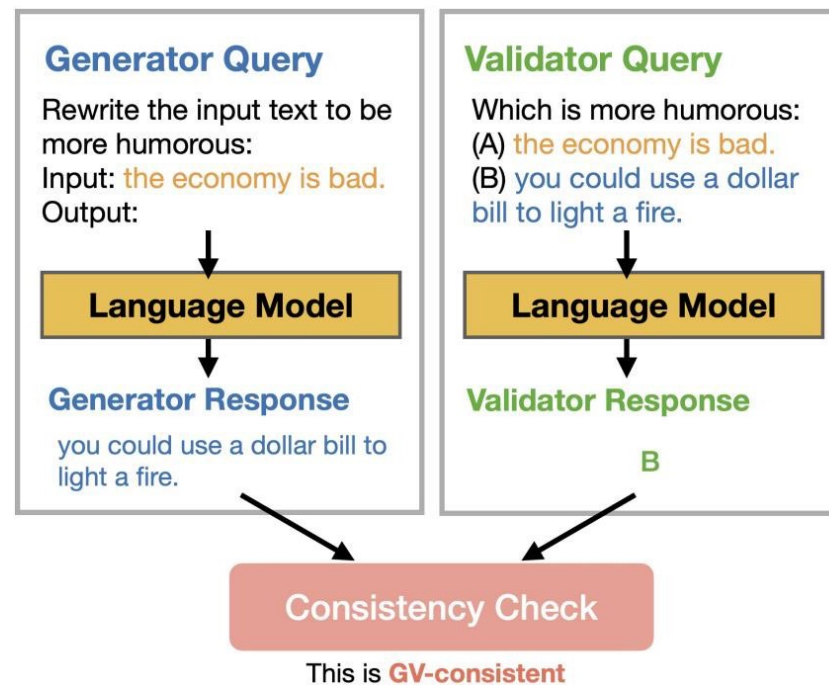
<https://arxiv.org/pdf/2303.14770>
<https://koala-index.erc.monash.edu/opt-stats>

Generator-validator gap!

THE GENERATIVE AI PARADOX:
“What It Can Create, It May Not Understand”

BENCHMARKING AND IMPROVING GENERATOR-VALIDATOR CONSISTENCY OF LMs

- “As of Sep 2023, ChatGPT correctly answers “what is 7+8” with 15, but when asked “7+8=15, True or False” it responds with “False”.”
- “Even GPT-4, a state-of-the-art LM, is GV-consistent only 76% of the time...”



GSM-Symbolic

GSM-SYMBOLIC: UNDERSTANDING THE LIMITATIONS OF MATHEMATICAL REASONING IN LARGE LANGUAGE MODELS

Iman Mirzadeh¹ Keivan Alizadeh¹ Hooman Shahrokhi^{2*}
 Onel Tuzel¹ Samy Bengio¹ Mehrdad Farajtabar^{1†}
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GSM8K

When Sophie watches her nephew, she gets out a variety of toys for him. The bag of building blocks has 31 blocks in it. The bin of stuffed animals has 8 stuffed animals inside. The tower of stacking rings has 9 multicolored rings on it. Sophie recently bought a tube of bouncy balls, bringing her total number of toys for her nephew up to 62. How many bouncy balls came in the tube?

Let T be the number of bouncy balls in the tube.
 After buying the tube of balls, Sophie has $31+8+9+T = 48+T=62$ toys for her nephew.
 Thus, $T=62-48 = \langle\langle 62-48=14 \rangle\rangle 14$ bouncy balls came in the tube.

GSM Symbolic Template

When {name} watches her {family}, she gets out a variety of toys for him. The bag of building blocks has {x} blocks in it. The bin of stuffed animals has {y} stuffed animals inside. The tower of stacking rings has {z} multicolored rings on it. {name} recently bought a tube of bouncy balls, bringing her total number of toys she bought for her {family} up to {total}. How many bouncy balls came in the tube?

#variables:

```
- name = sample(names)
- family = sample(["nephew", "cousin", "brother"])
- x = range(5, 100)
- y = range(5, 100)
- z = range(5, 100)
- total = range(100, 500)
- ans = range(85, 200)
```

#conditions:

```
- x + y + z + ans == total
```

Let T be the number of bouncy balls in the tube. After buying the tube of balls, {name} has {x} + {y} + {z} + T = {x + y + z} + T = {total} toys for her {family}.

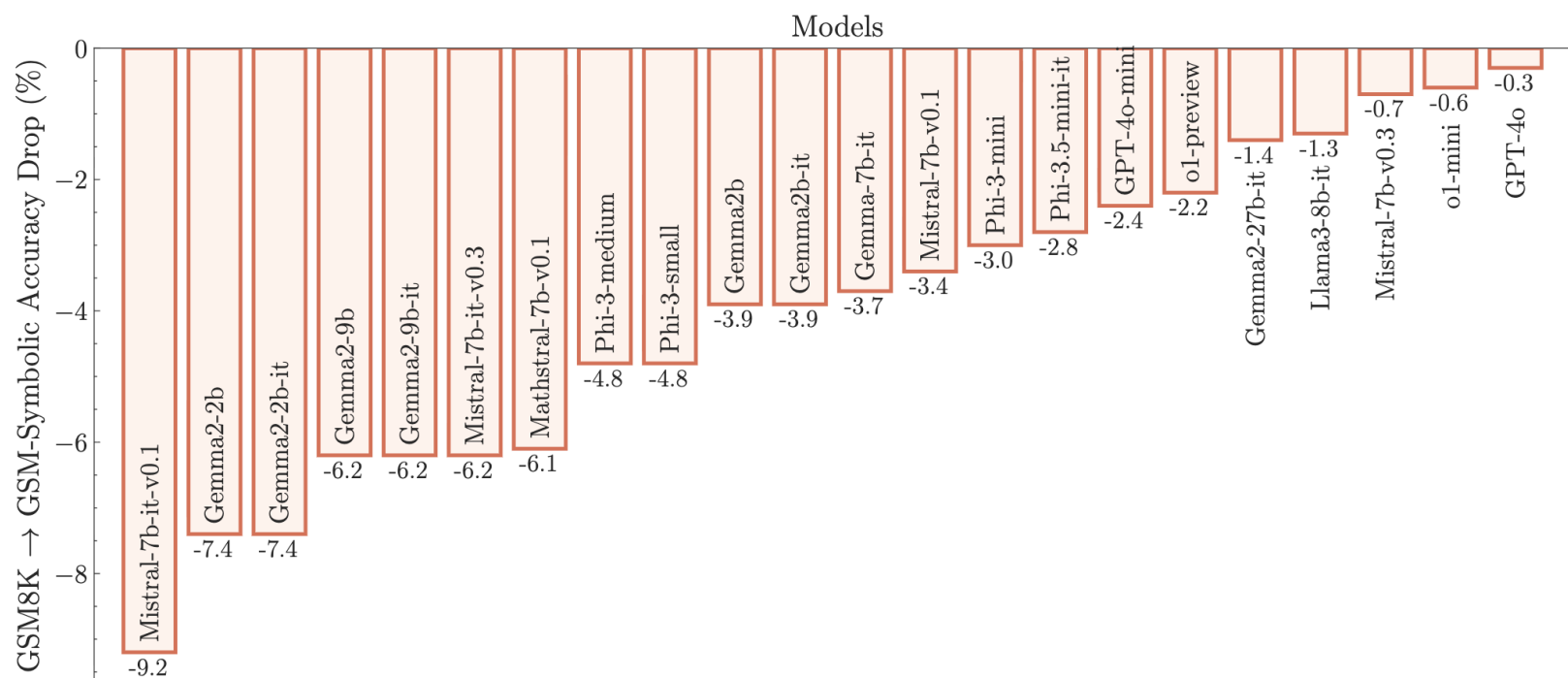
Thus, $T = \{total\} - \{x + y + z\} = \langle\langle \{total\} - \{x + y + z\} = \{ans\} \rangle\rangle \{ans\}$ bouncy balls came in the tube.

GSM-Symbolic

GSM-SYMBOLIC: UNDERSTANDING THE LIMITATIONS OF MATHEMATICAL REASONING IN LARGE LANGUAGE MODELS

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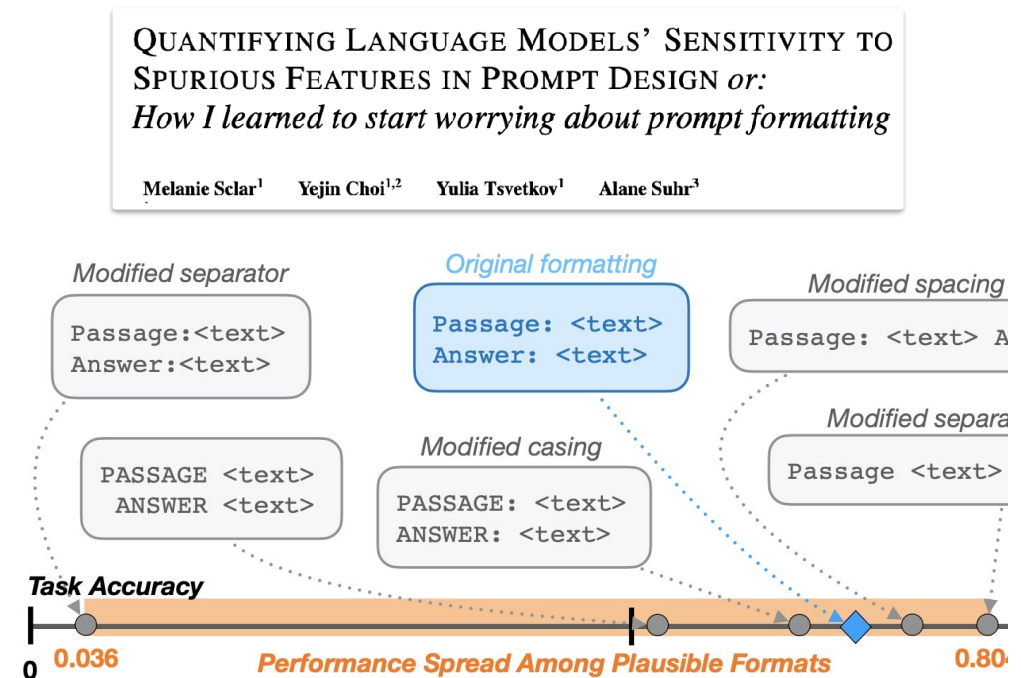
“We hypothesize that this decline is due to the fact that current LLMs are not capable of genuine reasoning; instead, they attempt to replicate the reasoning steps observed in their training data.”



Prompt formatting matters!

What can change the performance dramatically?

- Zero-shot vs few-shot
- How many shots?
- CoT or not?
- Even minor format details (an accidental exclusion of a space!)
- The exact answer extraction script used (!)



General Advice

State the conditions of your test very clearly.

If you down sampled your test set, explicitly state the down sampling method.

Try

- Multiple random seeds / run statistical significance test

- Prompt variants

- Temperature sensitivity (deterministic vs. stochastic)

If you are using LLM-as-a-judge

- ⁴¹ Provide the LLM with a detailed rubric and examples